

Electromagnetic Fields

Previous Year Questions • 2021–2026

GATE INSTRUMENTATION ENGINEERING

(IN)

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book is a complete compilation of the questions GATE has actually asked in Electromagnetic Fields (IN), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, tagged with its year, session and marks.

Each question carries an identifier that maps one-to-one onto its worked solution in the companion solutions volume, so you can move between practice and explanation without a lookup table. Answer keys close every section, and answers revised by the examining authority (MTA) are marked where they occur.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

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- AIR 27 — GATE (ECE)
- AIR 45 — GATE (IN)
- 5+ years of teaching experience



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> *"We built this book the way every aspirant deserves one to be built — organized strictly by unit and by year, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:





Easy



Medium



Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

How to Read the Unit Snapshot

Every unit opens with a **Unit Snapshot** box summarizing that unit's real question data:

Questions	the real total for this unit.
MCQ, MSQ, NAT	that same total, split by question type.
1M, 2M	how many of those questions carry 1 or 2 marks.
Descriptive	questions worth more than 2 marks, or whose marks value isn't recorded.
Avg Weightage	the average marks this unit has contributed per real GATE sitting, not a raw year-by-year total — a year GATE ran multiple real sessions (a genuine forenoon/afternoon split, not the same paper's Set A/B/C/D booklet variants) is divided across those sessions first, so a heavily multi-session year cannot silently outweigh a single-session one. The ~ marks it as a rounded figure, not an exact one.
Range	the lowest and highest that same per-sitting value has been, with the year each occurred.

Avg Weightage and Range are computed from **2010 onward** only: GATE dropped from a 150-mark to a 100-mark paper in 2009, but that year itself was a one-off transition (60 questions, not the 65-question format that became standard from 2010 on) — so 2010 is the first year fully comparable to modern papers, and marks-based figures from earlier years are left out of these two figures specifically (every other count on this page still includes every year). They also depend on knowing which GATE years ran more than one real sitting for a given branch — our own research into this is thorough for some branches and still a work in progress for others. Treat both figures as a **beta**, best-effort estimate: directionally useful, not exact.

MSQ questions specifically weren't introduced until 2021 — an older-year-heavy unit showing **MSQ: 0** reflects that real exam history, not a data gap.

Worked example

Unit Snapshot*

Stream: EC **Questions:** 45 **MCQ:** 30 **MSQ:** 3 **NAT:** 12
1M: 20 **2M:** 22 **Descriptive:** 3 **Avg Weightage** ~4M **Range:** 2M(2015)–6M(2021)

45 real questions total; on average this unit has been worth about 4 marks per real GATE sitting since 2010, ranging from as low as 2 (in 2015) to as high as 6 (in 2021).

A quick way to use this: a high **Avg Weightage** points to a consistently high-yield topic worth prioritizing; a wide **Range** usually means the topic's importance has shifted over the years, so it's worth checking *which* years drove that before deciding how much weight to give it today.

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Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

Contributors

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2	Magnetostatics	4

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UNIT

I

Electrostatics

Topics

1. Coulomb's Law and Superposition Principle
2. Electric Field, Electric Field Intensity calculation in various Configurations: Due to Point, Line, Plane and Spherical Charge Distributions
3. Equipotential Surfaces and Electric Dipoles
4. Electric Flux Density and Gauss's Law
5. Divergence and Electric Potential
6. Electric Field Effect of Dielectric Medium- a)Boundary conditions b)method of image charges
7. Capacitance of Simple Configurations

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Unit Snapshot*

Stream: IN	**Questions:** 8	**MCQ:** 6	**MSQ:** 0	**NAT:** 2
1M: 4	**2M:** 4	**Descriptive:** 0	**Avg Weightage** ~3M	**Range:** 1M(2021)–3M(2026)

**See preface for how Avg Weightage and Range are calculated.*

Q1.26.1 MCQ 2026 1M

An electric field has a potential of $V(x, y, z) = \sqrt{x^2 + y^2 + z^2}$ V .A charge of 1 coulomb placed at $(\hat{i} + \hat{j} + \hat{k})$ experiences a force of $\vec{F} = (a\hat{i} + b\hat{j} + c\hat{k})$ N The values of (a, b, c) are _____.

A. $a = \frac{1}{\sqrt{3}}, b = \frac{1}{\sqrt{3}}, c = \frac{1}{\sqrt{3}}$

B. $a = \frac{1}{3}, b = \frac{1}{3}, c = \frac{1}{3}$

C. $a = 0, b = 1, c = \frac{1}{\sqrt{3}}$

D. $a = 0, b = 1, c = \frac{1}{3}$

Q1.26.2 MCQ 2026 2M

An electric field in free space is $\vec{E} = (2x+5y+6z)\hat{i}+(5x+4y+10z)\hat{j}+(6x+10y+2z)\hat{k}$ kV/m The charge density is _____ C/m^3. (ε_0 is the permittivity of free space)

A. $8\varepsilon_0$ **B.** $10\varepsilon_0$ **C.** $6\varepsilon_0$ **D.** $4\varepsilon_0$

Q1.25.1 MCQ 2025 1M

An infinite sheet of uniform charge $\rho_s = 10\,\text{C/m}^2$ is placed on $z = 0$ plane. The medium surrounding the sheet has a relative permittivity of 10. The electric flux density, in C/m^2, at a point $P(0, 0, 5)$, is **Note:** $\hat{a}$, $\hat{b}$ and $\hat{c}$ and are unit vectors along the x, y, and z and directions, respectively.

A. $5\hat{c}$ **B.** $0.25\hat{c}$ **C.** $10\hat{c}$ **D.** $0.5\hat{c}$

Q1.25.2 NAT 2025 2M

An air filled parallel plate electrostatic actuator is shown in the figure. The area of each capacitor plate is $100\,\mu\text{m} \times 100\,\mu\text{m}$. The distance between the plates $d_0 = 1\,\mu$m when both the capacitor charge and spring restoring force are zero as shown in Figure (a). A linear spring of constant $k = 0.01$ N/m is connected to the movable plate. When charge is supplied to the capacitor using a current source, the top plate moves as shown in Figure (b). The magnitude of minimum charge (Q) required to momentarily close the gap between the plates is _____ $\times 10^{-14}$ C (rounded off to two decimal places). **Note:** Assume a full range of motion is possible for the top plate and there is no fringe capacitance. The permittivity of free space is $\varepsilon_0 = 8.854 \times 10^{-12}$ F/m and relative permittivity of air (ε_r) is 1.

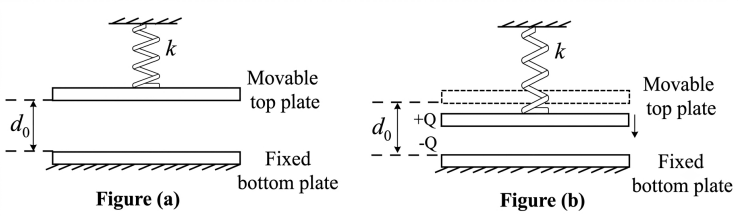


Fig. Q1.25.2

Q1.24.1 MCQ 2024 1M

The capacitance formed between two concentric spherical metal shells having radii x and y with $y > x$ is **Note:** ε is the permittivity of the medium between the shells.

A. $4\pi\varepsilon\left(\frac{xy}{y-x}\right)$ **B.** $4\pi\varepsilon\left(\frac{y^2}{y-x}\right)$

C. $4\pi\varepsilon\left(\frac{x^2}{y-x}\right)$ **D.** $4\pi\varepsilon\left(\frac{y^2-xy}{x}\right)$

Q1.24.2 MCQ 2024 2M

A wire of circular cross section with radius a is shown in the figure. The current density is given by $J = ks^2$, where k is a constant, s is the radial distance from the axis and $0 \le s \le a$. The total current I in the wire is

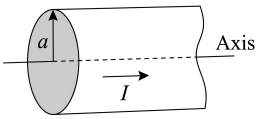


Fig. Q1.24.2

A. $\frac{\pi k a^4}{4}$ **B.** $\frac{\pi k a^4}{2}$ **C.** $\frac{2\pi k a^3}{3}$ **D.** $\frac{\pi k a^3}{2}$

Q1.22.1 NAT 2022 2M

A capacitor is constructed using two concentric spheres and air as the dielectric medium (permittivity of air = 8.854×10^{-12} F/m). The radii of the inner and outer spheres are $a = 10$ cm and $b = 15$ cm, respectively. The capacitance (in picofarads) is _____ (round off to 2 decimal places)

Q1.21.1 MCQ 2021 1M

An infinitely long line, with uniform positive charge density, lies along the z-axis. In cylindrical coordinates (r, ϕ, z), at any point $\vec{p}$ not on the z-axis, the direction of the electric field is

A. $\hat{z}$ **B.** $\hat{r}$ **C.** $\left(\frac{\hat{r}+\hat{z}}{\sqrt{2}}\right)$ **D.** $\hat{\phi}$

Answer Key

Note: MTA Marks To All awarded — see detailed solutions.

Q1.26.1 **MTA**	Q1.26.2 **A**	Q1.25.1 **A**
Q1.25.2 **4.00 to 4.40**	Q1.24.1 **A**	Q1.24.2 **B**
Q1.22.1 **33.3 to 33.4**	Q1.21.1 **B**	

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UNIT

II

Magnetostatics

Topics

1. Biot-Savart's Law
2. Electromagnetic Induction and Faraday's Law
3. Magnetic Field of a current carrying loop
4. Ampere's Law and Curl
5. Lorentz Force
6. Inductance and Magnetomotive Force
7. Self and Mutual Inductance of Simple Configurations
8. Reluctance and Magnetic Circuits
9. Magnetic Flux and Boundary Conditions for magnetic field

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Unit Snapshot*

Stream: IN	**Questions:** 6	**MCQ:** 2	**MSQ:** 0	**NAT:** 4
1M: 2	**2M:** 4	**Descriptive:** 0	**Avg Weightage** ~3M	**Range:** 1M(2025)–4M(2021)

**See preface for how Avg Weightage and Range are calculated.*

Q2.26.1 MCQ 2026 2M

A straight thin wire carrying a current of $I = 1\text{A}$ in the direction $\hat{n} = \frac{1}{\sqrt{6}}\hat{i} - \frac{2}{\sqrt{6}}\hat{j} + \frac{1}{\sqrt{6}}\hat{k}$ is placed inside a magnetic field of $\vec{B} = (2\hat{i} + 2\hat{k})$ T. The force per unit length of the wire is _____ N/m.

A. $2\sqrt{6}\hat{i} - 2\sqrt{6}\hat{k}$

B. $\sqrt{\frac{3}{2}}\hat{i} - \sqrt{\frac{3}{4}}\hat{j} + \sqrt{\frac{3}{4}}\hat{k}$

C. $\frac{4}{\sqrt{6}}\hat{i} - \frac{4}{\sqrt{6}}\hat{k}$

D. $\sqrt{\frac{3}{2}}\hat{i} + \sqrt{\frac{3}{4}}\hat{j} + \sqrt{\frac{3}{4}}\hat{k}$

Q2.25.1 MCQ 2025 1M

A 60 V DC source with an internal resistance $R_{\text{int}} = 0.5\,\Omega$ is connected through a switch to a pair of infinitely long rails separated by $l = 1$ m as shown in the figure. The rails are placed in a constant, uniform magnetic field of flux density $B = 0.5$ T, directed into the page. A conducting bar placed on these rails is free to move. At the instant of closing the switch, the force induced on the bar is **Note:** Assume there is no friction between the bar and the rails. The resistances of the conducting bar and the rails are zero.

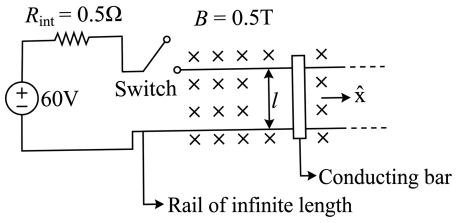


Fig. Q2.25.1

A. 60 N in the direction of $\hat{x}$

B. 60 N opposite to the direction of $\hat{x}$

C. 120 N in the direction of $\hat{x}$

D. 120 N opposite to the direction of $\hat{x}$

Q2.23.1 NAT 2023 1M

The force per unit length between two infinitely long parallel conductors, with a gap of 2 cm between them is 10 μN/m. When the gap is doubled, the force per unit length will be _____ μN/m (rounded off to one decimal place).

Q2.22.1 NAT 2022 2M

A conducting semi-circular loop of radius $R = 0.1$ m, with its diameter centered at the origin, rotates in the x-y plane about the origin with a constant angular velocity, $\omega = 20$ rad/s, as shown. A magnetic field of magnitude $B = 2$ T and normal to x - y plane exists in the region $x \geq 0$ as shown. If the loop has a resistance of 2 Ω and negligible inductance, the peak-to-peak current (in milliamperes) in the loop is _____ (round off to one decimal place)

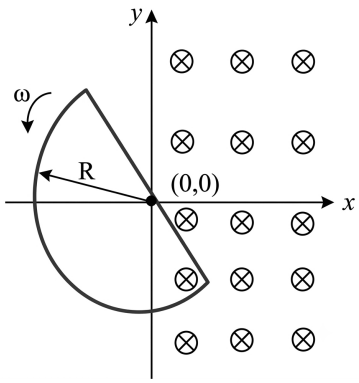


Fig. Q2.22.1

Q2.21.1 NAT 2021 2M

The figure below shows an electrically conductive bar of square cross-section resting on a plane surface. The bar of mass of 1 kg has a depth of 0.5 m along the y direction. The coefficient of friction between the bar and the surface is 0.1. Assume the acceleration due to gravity to be 10 m/s^2. The system faces a uniform flux density $B = -1\,\hat{z}$ T. At time $t = 0$, a current of 10 A is switched onto the bar and is maintained.When the bar has moved by 1m, its speed in metre per second is _____ (rounded off to one decimal place).

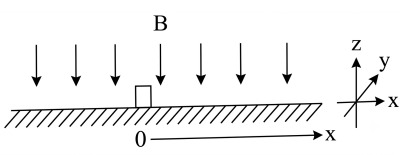


Fig. Q2.21.1

Q2.21.2 NAT | 2021 | 2M

A toroid made of CRGO has an inner diameter of 10 cm and an outer diameter of 14 cm. The thickness of the toroid is 2cm. 200 turns of copper wire is wound on the core. $\mu_0 = 4\pi \times 10^{-7}$ H/m and μ_R of CRGO is 3000. When a current of 5 mA flows through the winding, the flux density in the core in millitesla is _____.

Answer Key

Q2.26.1 C	Q2.25.1 A	Q2.23.1 5
Q2.22.1 200	Q2.21.1 2.7 to 2.9	Q2.21.2 10

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Also in this Series

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